

 B.TECH 1ST YEAR · 1ST SEMESTER

Engineering Physics

Complete Formula & Concept Revision Sheet



UNIT IV

Quantum Mechanics & Free Electron Theory

Last-Day Exam Revision · All Formulas · Key Concepts



Dual Nature of Matter



Heisenberg Uncertainty



Wave Function ψ



Schrödinger Equations



Particle in a Box



Classical Free Electron



Quantum Free Electron



Fermi-Dirac Distribution



Density of States



Electrical Conductivity



Covers ALL exam-critical formulas · Tables · Flashcards · Exam Tips · Quick Revision



1. DUAL NATURE OF MATTER – de Broglie Hypothesis

Core Formulas

$$\lambda = h / p$$

$$p = mv \rightarrow \lambda = h / mv$$

$$\lambda = h / \sqrt{2mE_k}$$

where $E_k = \frac{1}{2}mv^2$ is kinetic energy

For Accelerated Electron

$$\lambda = h / \sqrt{2meV}$$

V = accelerating voltage, $e = 1.6 \times 10^{-19}$ C

Key Concepts

- ▶ Every particle has an associated wave — **matter wave**
- ▶ λ is inversely proportional to momentum
- ▶ Massive objects $\rightarrow$ very small λ (not observable)
- ▶ Electrons $\rightarrow \lambda$ detectable (proved by diffraction)
- ▶ Wave-particle duality: particle AND wave behavior

Symbol Table

Symbol	Meaning	SI Unit
λ	de Broglie wavelength	m
h	Planck's constant	6.626×10^{-34} J·s
p	Momentum	kg·m/s
m	Mass	kg
E_k	Kinetic energy	J

2. HEISENBERG UNCERTAINTY PRINCIPLE

Fundamental Relations

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

$$\Delta E \cdot \Delta t \geq h / 4\pi$$

$$\Delta \theta \cdot \Delta L \geq h / 4\pi$$

Also written as: $\Delta x \cdot \Delta p_x \geq \hbar/2$ where $\hbar = h/2\pi$

Physical Significance

- ▶ Cannot simultaneously determine **exact position AND momentum**
- ▶ More precisely we know position $\rightarrow$ less we know momentum
- ▶ $\Delta E \cdot \Delta t$: energy of a state and its lifetime cannot both be exact
- ▶ This is a **fundamental property of nature**, NOT instrument error
- ▶ Proves classical orbits in atoms are impossible
- ▶ Explains **zero-point energy** (particle can't be perfectly at rest)

Pair	Uncertainty Relation	Physical Example
Position–Momentum	$\Delta x \cdot \Delta p_x \geq \hbar/4\pi$	Electron position in atom
Energy–Time	$\Delta E \cdot \Delta t \geq \hbar/4\pi$	Excited state lifetime
Angle–Angular Momentum	$\Delta \theta \cdot \Delta L \geq \hbar/4\pi$	Rotation of particles

3. WAVE FUNCTION (ψ)

Probability Interpretation

$$|\psi|^2 = \psi \cdot \psi^* = \text{Probability Density}$$

$$P(x) dx = |\psi(x)|^2 dx$$

Normalization Condition

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

Total probability of finding particle = 1 (certainty)

Properties of Valid ψ

Property	Meaning
Single-valued	One value at each point
Continuous	No abrupt breaks
Finite	Must not blow up to ∞
Normalizable	$\int \psi ^2 dx = 1$ possible
Square-integrable	$\int \psi ^2 dx$ must converge

ψ itself has NO direct physical meaning; $|\psi|^2$ does.

4. SCHRÖDINGER WAVE EQUATIONS

Time-Dependent Equation

$$i\hbar \partial\psi/\partial t = \hat{H}\psi$$

$$i\hbar \partial\psi/\partial t = [-\hbar^2/2m \cdot \partial^2/\partial x^2 + V] \psi$$

Governs how ψ evolves in time

Time-Independent Equation

$$\hat{H}\psi = E\psi$$

$$-\hbar^2/2m \cdot d^2\psi/dx^2 + V\psi = E\psi$$

$$d^2\psi/dx^2 + (2m/\hbar^2) (E - V)\psi = 0$$

Operator Form & Terms

Term	Symbol	Meaning
Hamiltonian	$\hat{H} = \hat{T} + \hat{V}$	Total energy operator
KE Operator	$\hat{T} = -\hbar^2/2m \cdot \nabla^2$	Kinetic energy
PE Operator	$\hat{V} = V(x)$	Potential energy
Momentum Op.	$\hat{p} = -i\hbar \partial/\partial x$	Momentum operator
$\hbar$ (h-bar)	$\hbar = h/2\pi$	Reduced Planck const

3D Time-Independent

$$\nabla^2 \psi + (2m/\hbar^2) (E - V)\psi = 0$$

$\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ (Laplacian)

5. PARTICLE IN A 1D INFINITE POTENTIAL WELL

Potential Setup

Region	Potential V(x)
$0 < x < L$ (inside)	$V = 0$
$x \leq 0$ or $x \geq L$ (outside)	$V = \infty$

Energy Eigenvalues

$$E_n = n^2 \pi^2 \hbar^2 / (2mL^2)$$

$$E_n = n^2 \hbar^2 / (8mL^2)$$

$n = 1, 2, 3, \dots$ (quantum number) | L = box length

Ground State (n=1) Energy

$$E_1 = \hbar^2 / (8mL^2) \rightarrow \text{Zero-Point Energy}$$

Wave Functions

$$\psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x/L)$$

Boundary conditions: $\psi(0) = 0$ and $\psi(L) = 0$

Key Observations

- Energy is **quantized** → only discrete values allowed
- Zero-point energy**: $E_1 \neq 0$ (particle cannot be at rest)
- $E_n \propto n^2$ → energy increases with square of quantum number
- $E_n \propto 1/L^2$ → smaller box, higher energy
- $E_n \propto 1/m$ → lighter particle, higher energy
- Wave function has n antinodes (humps)
- Probability density $|\psi|^2 \neq \text{uniform}$ (unlike classical)

Energy Level Ratios

n	E_n (multiples of E_1)	Nodes inside
1	E_1 (ground state)	0
2	$4E_1$	1
3	$9E_1$	2
4	$16E_1$	3
n	$n^2 E_1$	n-1

Probability Density

$$|\psi_n|^2 = (2/L) \cdot \sin^2(n\pi x/L)$$

Summary: All Quantum Formulas So Far

Concept	Formula	Key Note
de Broglie wavelength	$\lambda = h/p = h/mv$	All matter has wave nature
λ via kinetic energy	$\lambda = h/\sqrt{2mE_k}$	From $E = p^2/2m$
λ for accelerated e^-	$\lambda = h/\sqrt{2meV}$	V = voltage
Position-momentum uncertainty	$\Delta x \cdot \Delta p \geq \hbar/4\pi$	Fundamental limit
Energy-time uncertainty	$\Delta E \cdot \Delta t \geq \hbar/4\pi$	Excited state lifetime
Normalization	$\int \psi ^2 dx = 1$	Total probability = 1
TISE	$d^2\psi/dx^2 + (2m/\hbar^2)(E-V)\psi = 0$	Stationary states
Particle in box energy	$E_n = n^2 \hbar^2 / (8mL^2)$	Quantized levels
Particle in box ψ	$\psi_n = \sqrt{2/L} \cdot \sin(n\pi x/L)$	Standing waves

6. CLASSICAL FREE ELECTRON THEORY (Drude–Lorentz Model)

Assumptions

- Free electrons move randomly (like gas molecules)
- Electron–electron and electron–ion interactions neglected
- Electric field accelerates electrons → drift velocity
- Collisions randomize velocity (relaxation time τ)
- Obeys Maxwell-Boltzmann statistics

Drift Velocity

$$v_d = eE\tau / m$$

E = applied electric field, τ = relaxation time

Current Density

$$J = nev_d = ne^2E\tau / m$$

n = number of free electrons per unit volume

Ohm's Law & Conductivity

$$J = \sigma E$$

$$\sigma = ne^2\tau / m$$

$$\rho = 1/\sigma = m / (ne^2\tau)$$

More Formulas

Quantity	Formula
Mobility (μ)	$\mu = v_d/E = e\tau/m$
Conductivity	$\sigma = ne\mu$
Mean free path	$\lambda = v_{th} \cdot \tau$
Thermal velocity	$v_{th} = \sqrt{3k_B T/m}$
Resistance	$R = \rho L/A$

Merits & Demerits

✓ Merits	✗ Demerits
Explains Ohm's law	Cannot explain specific heat
Explains electrical conductivity	Incorrect temp dependence of σ
Explains thermal conductivity	Fails for semiconductors
Wiedemann-Franz law	No quantum effects considered
Simple and intuitive	Over-estimates electron contribution

Wiedemann–Franz Law

$$K / \sigma = LT$$

K = thermal conductivity, L = Lorentz number, T = temperature

$$L = \pi^2 k_B^2 / (3e^2) = 2.45 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$$

- Ratio K/σ is proportional to temperature T
- Good electrical conductors are also good thermal conductors
- Verified experimentally for metals at room temperature
- L is the **Lorentz number** (universal constant)

7. QUANTUM FREE ELECTRON THEORY (Sommerfeld Model)

Key Assumptions

- ▶ Electrons obey **Fermi–Dirac statistics** (not Maxwell-Boltzmann)
- ▶ Electrons treated as **quantum particles** in a potential well
- ▶ Pauli exclusion principle: max 2 electrons per energy state
- ▶ Only electrons near Fermi level participate in conduction
- ▶ Electron–electron interactions still neglected

Fermi Energy

$$E_F = (\hbar^2 / 2m) (3\pi^2 n)^{2/3}$$

n = electron density (electrons/m³)

$$E_F = (\hbar^2 / 8m) (3n/\pi)^{2/3}$$

Classical vs Quantum Comparison

Property	Classical	Quantum
Statistics	Maxwell-Boltzmann	Fermi-Dirac
Energy at 0K	All zero	Up to E_F
Specific heat	$3/2 nk_B T$ (wrong)	$\sim (\pi^2/2)(k_B T/E_F) \cdot nk_B$
Conductors	All electrons	Only near Fermi level
Exclusion principle	Not applied	Applied (Pauli)
σ vs T	$\sigma \propto 1/T$	$\sigma \propto 1/T$ (metals, correctly)

8. FERMI–DIRAC DISTRIBUTION FUNCTION

Distribution Function

$$f(E) = 1 / [\exp((E - E_F)/k_B T) + 1]$$

$f(E)$ = probability that state with energy E is occupied

Special Cases

Condition	Result
$T = 0 \text{ K}, E < E_F$	$f(E) = 1$ (fully occupied)
$T = 0 \text{ K}, E > E_F$	$f(E) = 0$ (empty)
$E = E_F$ (any T)	$f(E) = 1/2$
$E \gg E_F$	$f(E) \rightarrow \text{Maxwell-Boltzmann}$

Graph Characteristics

- ▶ At $T = 0 \text{ K}$: **step function** (sharp cutoff at E_F)
- ▶ At $T > 0 \text{ K}$: curve **smears out** near E_F
- ▶ $f(E_F) = 0.5$ always, at all temperatures
- ▶ Width of smearing $\approx k_B T$
- ▶ Higher $T \rightarrow$ more electrons above E_F

Fermi Temperature

$$T_F = E_F / k_B$$

Typical metals: $T_F \sim 10,000 \text{ K}$ (very high)

Fermi Velocity

$$v_F = \sqrt{2E_F/m}$$

9. DENSITY OF STATES (DOS)

Definition & Formula

$$g(E) = \frac{(4\pi/h^3) (2m)^{3/2}}{\sqrt{E}}$$

$$g(E) = \frac{(3n/2) \cdot E^{1/2}}{E_F^{3/2}}$$

$g(E) dE$ = number of quantum states in energy range E to $E+dE$

Graph Interpretation

- $g(E) \propto \sqrt{E}$ — **parabolic** increase with energy
- At $E = 0$: $g(E) = 0$
- As E increases: more states available
- At $T = 0K$: all states up to E_F are filled

Number of Electrons

$$n = \int_0^{E_F} g(E) f(E) dE$$

At $T = 0K$: $n = \int_0^{E_F} g(E) dE$

Energy Dependence Summary

Quantity	Energy Dependence
DOS $g(E)$	$\propto E^{1/2}$
Fermi-Dirac $f(E)$	Step function at 0K
Occupied states	$g(E) \cdot f(E)$
Fermi energy E_F	$\propto n^{2/3}$

10. ELECTRICAL CONDUCTIVITY – QUANTUM MODEL

Quantum Formula

$$\sigma = ne^2 \tau / m$$

Same form as classical, but τ now depends on Fermi velocity

$$\sigma = ne^2 \tau_F / m^*$$

τ_F = relaxation time at Fermi level, m^* = effective mass

Mean Free Path at Fermi Level

$$\lambda_F = v_F \cdot \tau_F$$

Key Points

- Only electrons **near Fermi level** contribute to conductivity
- Relaxation time τ = average time between collisions
- $\sigma \propto \tau \rightarrow$ higher $\tau \rightarrow$ better conductor
- In metals: σ decreases with T (τ decreases)
- In semiconductors: σ increases with T (n increases)
- Effective mass m^* accounts for crystal potential

Temperature Dependence

Material	σ vs T	Reason
Metals	$\sigma \propto 1/T$	τ decreases (more vibrations)
Semiconductors	σ increases with T	More carriers excited
Superconductors	$\sigma \rightarrow \infty$ below T_c	Zero resistance

Important Constants

Constant	Symbol	Value
Planck's constant	h	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
Reduced Planck's	$\hbar = h/2\pi$	$1.055 \times 10^{-34} \text{ J}\cdot\text{s}$
Electron mass	m	$9.11 \times 10^{-31} \text{ kg}$
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Boltzmann constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of light	c	$3 \times 10^8 \text{ m/s}$



FLASHCARDS (25 Cards – Exam-Oriented)

Q: What is de Broglie wavelength?

A: $\lambda = h/p = h/mv$

Q: λ for electron accelerated through voltage V?

A: $\lambda = h/\sqrt{2meV}$

Q: State Heisenberg's uncertainty principle

A: $\Delta x \cdot \Delta p \geq h/4\pi$; $\Delta E \cdot \Delta t \geq h/4\pi$

Q: What does $|\psi|^2$ represent?

A: Probability density of finding a particle

Q: Normalization condition of wave function?

A: $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$

Q: What is the time-independent Schrödinger equation?

A: $d^2\psi/dx^2 + (2m/\hbar^2)(E-V)\psi = 0$

Q: Energy of nth state in particle in a box?

A: $E_n = n^2 h^2 / (8mL^2)$

Q: Wave function in 1D box?

A: $\psi_n = \sqrt{2/L} \cdot \sin(n\pi x/L)$

Q: What is zero-point energy?

A: $E_1 = h^2 / (8mL^2)$ – minimum energy even at 0K

Q: Drift velocity in classical theory?

A: $v_d = eE\tau/m$

Q: Expression for electrical conductivity?

A: $\sigma = ne^2\tau/m$

Q: What is electron mobility?

A: $\mu = v_d/E = e\tau/m$

Q: What is Fermi energy?

A: Highest occupied energy level at T = 0K

Q: Fermi-Dirac distribution function?

A: $f(E) = 1/[\exp((E-E_F)/k_B T) + 1]$

Q: $f(E)$ at $E = E_F$?

A: $f(E_F) = 1/2$, at any temperature

Q: Density of states formula?

A: $g(E) = (4\pi/h^3) (2m)^{3/2} \sqrt{E} \propto \sqrt{E}$

Q: Which statistics do quantum electrons obey?

A: Fermi-Dirac statistics (Pauli exclusion applies)

Q: What is Wiedemann-Franz law?

A: $K/\sigma = LT$ (L = Lorentz number)

Q: Lorentz number value?

A: $L = 2.45 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$

Q: Energy dependence of DOS?

A: $g(E) \propto E^{1/2}$ (parabolic)

Q: Fermi energy formula?

A: $E_F = (\hbar^2/2m) (3\pi^2 n)^{2/3}$

Q: What is $\hbar$ (h-bar)?

A: $\hbar = h/2\pi = 1.055 \times 10^{-34} \text{ J}\cdot\text{s}$

Q: Condition for valid wave function?

A: Single-valued, continuous, finite, normalizable

Q: How does σ vary with temperature in metals?

A: σ decreases ($\sigma \propto 1/T$), because τ decreases

Q: What is the Hamiltonian operator $\hat{H}$?

A: $\hat{H} = -\hbar^2/2m \cdot \nabla^2 + V(r)$ (Total energy operator)

⚡ EXAM TIPS – Avoid These Mistakes!

✗ Mistake 1: Wrong Uncertainty Formula

Wrong: $\Delta x \cdot \Delta p \geq \hbar/2\pi$

✓ Correct: $\Delta x \cdot \Delta p \geq \hbar/4\pi = \hbar/2$

Remember: 4π in denominator, NOT 2π

✗ Mistake 2: Forgetting n^2 in Box Energy

Wrong: $E_n = \hbar^2/(8mL^2)$

✓ Correct: $E_n = n^2\hbar^2/(8mL^2)$

n^2 is crucial — energy scales with n squared

✗ Mistake 3: ψ vs $|\psi|^2$ Confusion

ψ = wave function (complex number, no direct meaning)

✓ $|\psi|^2$ = probability density (measurable, always real ≥ 0)

Always use $|\psi|^2$ for probability questions

✗ Mistake 4: Normalization Error

Don't forget: The integral of $|\psi|^2$ must equal 1

✓ Check: $\int |\psi|^2 dx = 1$ after finding ψ_n

Normalization constant $A = \sqrt{2/L}$ for box

✗ Mistake 5: h vs $\hbar$

$h = 6.626 \times 10^{-34}$ J-s (Planck's constant)

✓ $\hbar = h/2\pi = 1.055 \times 10^{-34}$ J-s

TISE uses $\hbar^2$; box energy uses h^2

✗ Mistake 6: Wrong Boundary Conditions

For box: $\psi(0) = 0$ AND $\psi(L) = 0$

✓ This gives: $k = n\pi/L \rightarrow E_n = n^2\pi^2\hbar^2/2mL^2$

$n = 1, 2, 3...$ (not 0, as $E_0 = 0$ is forbidden)

✗ Mistake 7: Classical vs Quantum Statistics

Classical $\rightarrow$ Maxwell-Boltzmann statistics

✓ Quantum electrons $\rightarrow$ Fermi-Dirac statistics

Never apply MB to free electrons in metals

✗ Mistake 8: Fermi-Dirac at $E = E_F$

Wrong: $f(E_F) = 0$ or $f(E_F) = 1$

✓ Correct: $f(E_F) = 1/2$, at any temperature T

This is a fixed result — always 0.5 at Fermi level

✗ Mistake 9: Confusing σ Dependence

Metals: σ decreases with T (τ decreases)

Semiconductors: σ increases with T (more carriers)

✓ Know which material type is being asked about

✗ Mistake 10: DOS Energy Dependence

Wrong: $g(E) \propto E$ or $g(E) = \text{constant}$

✓ Correct: $g(E) \propto E^{1/2}$ (square root)

Parabolic shape — not linear, not constant

💡 Pro Tip: Unit Checking

Always verify units in numerical problems:

λ in meters (m), E in Joules (J) or eV

1 eV = 1.6×10^{-19} J — conversion is critical!

Convert eV $\rightarrow$ J before substituting in formulas

💡 Pro Tip: Memorization Strategy

de Broglie: $\lambda = h/p \rightarrow$ "wavelength = Planck/momentum"

Box energy: $E_n = n^2\hbar^2/8mL^2 \rightarrow$ "n-squared, h-squared, 8mL²"

Fermi-Dirac: denominator = $e^{...} + 1 \rightarrow$ "+1 makes it quantum"

📄 IMPORTANT DERIVATION STEPS (Summary Only)

Derivation	Key Steps	Final Result
Particle in Box Energy	Apply TISE with $V=0 \rightarrow \sin/\cos$ solutions $\rightarrow$ BC gives $kL=n\pi \rightarrow k=n\pi/L \rightarrow E=\hbar^2k^2/2m$	$E_n = n^2\hbar^2/8mL^2$
de Broglie for electron	$E = eV \rightarrow \frac{1}{2}mv^2 = eV \rightarrow v = \sqrt{2eV/m} \rightarrow p = mv \rightarrow \lambda = h/p$	$\lambda = h/\sqrt{2meV}$
Conductivity (Classical)	$F = eE \rightarrow a = eE/m \rightarrow v_d = a\tau = eE\tau/m \rightarrow J = nev_d = ne^2E\tau/m \rightarrow J = \sigma E$	$\sigma = ne^2\tau/m$

MASTER FORMULA REFERENCE – No Explanations, Pure Formulas

DE BROGLIE

$$\lambda = h/p$$

$$\lambda = h/mv$$

$$\lambda = h/\sqrt{(2mE_k)}$$

$$\lambda = h/\sqrt{(2meV)}$$

HEISENBERG

$$\Delta x \cdot \Delta p \geq h/4\pi$$

$$\Delta E \cdot \Delta t \geq h/4\pi$$

$$\Delta x \cdot \Delta p_x \geq h/2$$

WAVE FUNCTION

$$|\psi|^2 = \text{prob. density}$$

$$\int |\psi|^2 dx = 1$$

$$P = \int_a^b |\psi|^2 dx$$

SCHRÖDINGER (TISE)

$$\hat{H}\psi = E\psi$$

$$d^2\psi/dx^2 + (2m/\hbar^2)(E-V)\psi = 0$$

$$i\hbar\partial\psi/\partial t = \hat{H}\psi \quad (\text{TDSE})$$

PARTICLE IN BOX

$$E_n = n^2 h^2 / 8mL^2$$

$$E_n = n^2 \pi^2 \hbar^2 / 2mL^2$$

$$\psi_n = \sqrt{(2/L)} \cdot \sin(n\pi x/L)$$

$$E_1 = h^2 / 8mL^2 \quad (\text{ZPE})$$

OPERATORS

$$\hat{H} = -\hbar^2/2m \cdot \nabla^2 + V$$

$$\hat{p} = -i\hbar \partial/\partial x$$

$$\hbar = h/2\pi$$

CLASSICAL FREE E⁻

$$v_d = eE\tau/m$$

$$J = nev_d$$

$$\sigma = ne^2\tau/m$$

$$\mu = e\tau/m$$

$$\rho = m/ne^2\tau$$

QUANTUM THEORY

$$E_F = \hbar^2 (3\pi^2 n)^{2/3} / 2m$$

$$f(E) = 1/[e^{(E-E_F)/k_B T} + 1]$$

$$g(E) \propto \sqrt{E}$$

$$v_F = \sqrt{(2E_F/m)}$$

WIEDEMANN-FRANZ

$$K/\sigma = LT$$

$$L = 2.45 \times 10^{-8} \text{ WOK}^{-2}$$

COMPLETE FORMULA TABLE

#	Topic	Formula	Variables
1	de Broglie	$\lambda = h/p$	h =Planck, p =momentum
2	de Broglie (KE)	$\lambda = h/\sqrt{(2mE_k)}$	E_k =kinetic energy
3	de Broglie (voltage)	$\lambda = h/\sqrt{(2meV)}$	V =voltage, e =charge
4	Uncertainty (pos-mom)	$\Delta x \cdot \Delta p \geq h/4\pi$	Δ =uncertainty
5	Uncertainty (energy-time)	$\Delta E \cdot \Delta t \geq h/4\pi$	—
6	Normalization	$\int \psi ^2 dx = 1$	ψ =wave function
7	TISE	$d^2\psi/dx^2 + (2m/\hbar^2)(E-V)\psi = 0$	$\hbar=h/2\pi$, V =potential
8	Box energy	$E_n = n^2 h^2 / 8mL^2$	n =quantum no., L =length
9	Box wave fn	$\psi_n = \sqrt{(2/L)} \cdot \sin(n\pi x/L)$	—
10	Drift velocity	$v_d = eE\tau/m$	τ =relaxation time
11	Conductivity	$\sigma = ne^2\tau/m$	n =electron density
12	Mobility	$\mu = e\tau/m = \sigma/ne$	—
13	Fermi energy	$E_F = (\hbar^2/2m) (3\pi^2 n)^{2/3}$	—
14	Fermi-Dirac	$f(E) = 1/[e^{(E-E_F)/k_B T} + 1]$	k_B =Boltzmann const
15	DOS	$g(E) = (4\pi/\hbar^3) (2m)^{3/2} \sqrt{E}$	—
16	Fermi velocity	$v_F = \sqrt{(2E_F/m)}$	—
17	Wiedemann-Franz	$K/\sigma = LT$	L =Lorentz, K =thermal cond.

